

Rishabh Dev Singh

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SUMMARY

Data Engineer with 3+ years of experience building AWS batch and streaming pipelines using PySpark, SQL, Kafka/MSK, and Kinesis. At Moody's, driving regulatory reports automation with measurable SLA, cost, and data quality improvements.

WORK EXPERIENCE

Moody's Ratings Technology Aug 2023 – Present

Data Engineer II

- Own and operate a mission-critical regulatory compliance pipeline with complex multi-stage transformations across 20+ Glue Scripts — submits daily reports to ESMA and FCA regulators, receives feedback, and maintains a <3% error rate under contractual SLA obligations across 3 LOBs
- Coordinated between data engineering and application development teams to design API data contracts and validate response schemas for downstream consumers.
- Led 3 interns on agent-based POCs — owned design reviews and drove data enrichment strategies shaped by regulatory data complexity.
- Created and delivered 3 knowledge sessions (20+ attendees) to drive adoption of AI-assisted development tooling across the team, covering Claude usage, RAG workflows, and token optimization techniques.

Cognizant Dec 2020 – Apr 2021

Programmer Analyst

- Completed hands-on Salesforce platform training as part of the analyst onboarding program.

EDUCATION

IIT Guwahati Jul 2021 – Jul 2023

M.Tech in Computer Science & Engineering

IET Indore Jul 2016 – Jul 2020

B.Tech, Information Technology

PROJECTS

Regulatory Reporting Redesign Aug 2023 – Present | Moody's

- A complex daily compliance reporting system submitting credit rating data to regulators before public release — migrated from legacy on-premise to AWS cloud. Any delay or error carries direct regulatory and reputational risk for Moody's.
- Achieved 85% reduction in submission errors (~20% to ~3%) by migrating off a legacy route unable to handle multi-stage transformation complexity; sustaining under 1% error rate in stable operation.
- Automated PySpark ETL optimization (partitioning, caching, I/O tuning), cutting per-run execution time by 21% (from 70 to 55 minutes) and eliminating 100% of recurring SLA breaches.
- Coordinated with API, database, and DevOps teams across 5+ cross-functional groups to align on data contracts, resolve integration issues, and deliver shared pipeline dependency agreements.

Agent-Based Prediction POC 2024 – 2025 | Moody's

- Built a regulatory submission failure predictor using engineered features and XGBoost, then re-architected it as a multi-agent LLM pipeline with structured prompt orchestration, improving prediction accuracy and reducing inference latency.
- Created 10+ custom Claude Code skills and hooks to automate Glue job debugging, Athena querying, and team workflows — reducing developer context-switching by ~30%.
- Designed a RAG pipeline over 50+ internal regulatory documents (ESMA/FCA specs, runbooks), cutting average knowledge lookup time from hours to seconds for the team.

Reliability Framework for Streaming Pipeline Individual

- Accomplished zero data loss in adversarial testing by designing and deploying a real-time streaming platform (API Gateway → Kafka/MSK → Kinesis) within a secured VPC, handling 10K+ events/minute.
- Delivered 100% pass rate across all failure mode tests — validated resilience under duplicate, late, out-of-order, and spike-traffic conditions; confirmed idempotency and DLQ handling.
- Implemented controlled replay and backfill using Kafka consumer group offsets, enabling safe offset resets for late-arriving events.

SKILLS

Languages	Python, SQL, Bash/Shell
Data Engineering	Apache Spark, PySpark Optimization (partitioning, shuffle, caching), ETL/ELT, CDC, Pipeline Design
AWS	S3, Glue, Lambda, EMR, Redshift, Kinesis Data Streams, Firehose, MSK (Kafka), IAM, VPC, CloudFormation, Step Functions, RDS/PostgreSQL
DevOps	Git, CI/CD (GitHub Actions), Docker, Infrastructure-as-Code
AI/ML	LLM Integration, RAG Pipelines, Prompt Engineering, Multi-Agent Orchestration, XGBoost

ACHIEVEMENTS

- 2x Recognition & Reward awardee at Moody's for technical excellence in pipeline automation and reporting reliability.
- Solved 500+ problems on LeetCode, demonstrating strong DSA and problem-solving fundamentals.